A-Block: Clifford Neural Score Matching for Multi-Well Conformational Dynamics and Active Particle Sampling

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Abstract

Sampling multi-well free energy surfaces and navigating metastable conformational transition states are central challenges in molecular machine learning. Standard score-guided Langevin samplers and diffusion models operate over unconstrained Euclidean gradients, frequently suffering from kinetic trapping in deep potential wells. In this work, we present *A-Block*: a Clifford Neural Score Matching framework that unifies Geometric Algebra ($\mathrm{Cl}(2,0)$), score-based stochastic differential equations, and active matter Langevin hydrodynamics for biomolecular sampling. We parameterize non-equilibrium potential fields via a neural network trained through Denoising Score Matching (DSM), augmented with coordinate-free multivector rotor-dilator dynamics $(\sigma_i I - \kappa_i)\nabla V_\theta(\mathbf{x}_i)$. We evaluate A-Block on a canonical multi-basin conformational landscape with three distinct metastable states (Basins A, B, C), benchmarking against unconditional Brownian dynamics, standard Langevin score matching, and an ablation on rotor-only dynamics ($\kappa = 0$). Our empirical results demonstrate that A-Block achieves the lowest Free Energy Surface reconstruction error (RMSE $= 1.059$ kcal/mol vs 1.476 kcal/mol for standard Langevin and 2.572 kcal/mol for Brownian dynamics) while maintaining linear per-frame computational complexity $O(N)$. We further demonstrate programmable spatial particulate control across non-convex geometric attractors, validating decoupled linear scaling empirically. This work establishes a geometric and physical foundation for enhanced molecular conformation sampling and active matter design.

1 Introduction

Characterizing complex free-energy landscapes, predicting metastable transition paths, and generating diverse biomolecular conformations are central objectives of molecular machine learning [1–3]. While score-based generative modeling and geometric deep learning have enabled rapid structural synthesis [4, 5], sampling from multi-well non-equilibrium distributions remains computationally challenging. In physical systems with high free-energy barriers between metastable states—such as multi-state signaling proteins undergoing conformational interchange [6]—standard Langevin samplers frequently suffer from kinetic trapping, requiring extensive simulation times to achieve representative equilibrium coverage.

Concurrently, physical active matter and non-reversible stochastic differential equations demonstrate how non-equilibrium forces and localized active circulation can drive spontaneous state transitions and overcome geometric barriers [7–9]. Integrating active rotor dynamics with score-based neural fields presents a promising direction: non-gradient transverse circulation along score isolines can

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accelerate barrier crossing and saddle-point exploration without degrading the underlying energy landscape.

In this paper, we introduce **A-Block** (Algebraic Block), a Clifford Neural Score Matching architecture for multi-well conformational dynamics and spatial particulate control (Figure 1). Rather than viewing kinetic trapping solely as a computational failure to be avoided, we frame it as a physical phenomenon to be programmatically governed: the multivector rotor (σI) drives particles across high-energy barriers to escape unwanted metastable traps, while the dilator ($-\kappa$) provides the contractive descent necessary to stabilize ensembles into target states—whether capturing equilibrium conformational basins or patterning arbitrary particulate attractors. Our contributions are threefold: (1) **Clifford Neural Score Architecture:** We formulate a neural score model $V_\theta(\mathbf{x})$ trained via Denoising Score Matching, coupled with a 2D Geometric Algebra ($\mathrm{Cl}(2,0)$) multivector rotor-dilator operator $(\sigma I - \kappa)$ that decouples naturally across backbone dihedral hinges without exponential basis explosion; (2) **Decoupled Point-Measure Complexity:** We prove that transitioning from pairwise proximity graphs ($O(N^2)$) to decoupled active density measures reduces per-step sampling cost to $O(N)$; and (3) **Empirical Grounding & Metric Analysis:** We establish that Free Energy Surface (FES) RMSE is the true indicator of thermodynamic fidelity over unphysical Wasserstein dispersion, validating A-Block against standard Langevin sampling, unconditional diffusion, and a rotor-only ablation ($\kappa = 0$).

2 The A-Block Architecture: Clifford Neural Score Matching

An *A-Block* models a modular ensemble of N particulate conformational states $\mathbf{x}_i \in \Omega \subset \mathbb{R}^2$ evolving under a learned score field.

2.1 Neural Potential Parameterization and Denoising Score Matching

Let $\mathbf{x} \in \mathbb{R}^2$ represent internal coordinates (e.g., conformational dihedral angles) of a molecular system governed by a multi-well potential $V(\mathbf{x})$. We model the target distribution $p(\mathbf{x}) \propto \exp(-V(\mathbf{x})/k_B T)$ using a parameterized neural scalar potential $V_\theta(\mathbf{x}) : \mathbb{R}^2 \to \mathbb{R}$.

The network is trained using Vincent–Song Denoising Score Matching (DSM) [5]:

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{\mathbf{x}\sim p_{\text{data}},\, \sigma\sim p(\sigma),\, \boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\sigma^2 \left\| -\nabla_{\tilde{\mathbf{x}}} V_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \right\|^2\right], \tag{1}$$

where $\tilde{\mathbf{x}} = \mathbf{x} + \sigma\boldsymbol{\epsilon}$ is the perturbed state, and the analytical score target is $\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) = -\frac{\boldsymbol{\epsilon}}{\sigma}$.

2.2 Geometric Algebra Formulation of Rotor-Dilator Dynamics ($\mathrm{Cl}(2,0)$)

Standard score matching drives particles strictly along the negative gradient $-\nabla V_\theta(\mathbf{x})$. To enable transversal circulation along score contours and escape local energy traps, we utilize the 2D Geometric Algebra ($\mathrm{Cl}(2,0)$) with orthonormal basis $\{e_1, e_2\}$ and unit pseudoscalar bivector $I = e_1 e_2$ ($I^2 = -1$). While Geometric Algebra over D dimensions generally exhibits an exponential basis expansion (2^D), high-dimensional biomolecular backbones naturally decouple into a direct sum of independent planar bivectors $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$ parameterized by local (ϕ_k, ψ_k) Ramachandran dihedral hinges, scaling as $4K$ rather than 2^{2K}.

We define the multivector score update operator as a combined *rotor-dilator*:

$$\mathbf{f}_i(\mathbf{x}_i) = (\sigma_i I - \kappa_i)\nabla V_\theta(\mathbf{x}_i), \tag{2}$$

where $I\nabla V_\theta = (-\partial_y V_\theta, \partial_x V_\theta)^T$ generates coordinate-free circulation orthogonal to the gradient, $\sigma_i \in [-1.5, 1.5]$ represents localized rotor chirality predicted by the network, and $\kappa_i = \mathrm{softplus}(\cdot) + 0.1 \in [0.1, \infty)$ represents the restoring dilator force toward potential minima.

Biophysically, standard Langevin descent resembles dropping a bead into a basin, where it rolls straight down and remains kinetically trapped. The Clifford rotor σI acts analogously to swirling liquid in a vessel: orthogonal circulation drives trajectories along equipotential contours, imparting non-conservative momentum to spill over steric barriers into neighboring basins. The restoring dilator $-\kappa$ provides energetic relaxation, ensuring the particle settles into the newly discovered minimum.

2.3 Active Overdamped Langevin Hydrodynamics

Each conformational particle evolves according to the overdamped stochastic differential equation (SDE):

$$\gamma d\mathbf{x}_i = \mathbf{f}_i(\mathbf{x}_i)dt + \sqrt{2\gamma k_B T} d\mathbf{W}_i(t), \tag{3}$$

where γ is the friction coefficient, $k_B T$ is thermal energy, and $d\mathbf{W}_i(t)$ represents standard 2D Brownian motion.

3 Complexity Analysis

In spatial computing and molecular structure generation, maintaining relational graph topologies $G_\varepsilon = (V, E_\varepsilon)$ requires quadratic distance computations ($O(N^2)$). A key structural property of A-Block is the transition from combinatorial graphs to continuous empirical point measures $\mu_t \in \mathcal{M}_+(\Omega)$:

$$G_\varepsilon = (V, E_\varepsilon) \xrightarrow{\mathcal{T}} \mu_t = \frac{1}{N}\sum_{i=1}^{N} \delta_{\mathbf{x}_i(t)}. \tag{4}$$

Proposition 1 (Linear Complexity across Decoupled Planar Rotors)**.** *Evaluating pairwise distances across N particles in a metric proximity graph G_ε incurs $\frac{N(N-1)}{2}$ operations ($O(N^2)$). Decoupling relational edges via empirical measure transformation $\mathcal{T}$ and localized Clifford score updates $\mathbf{f}_i(\mathbf{x}_i)$ reduces per-frame sampling complexity to strictly $O(N)$. Crucially, this linear scaling holds true even when decoupling higher-dimensional conformational spaces into independent planar 2D rotors $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$, scaling as $O(K \cdot N)$ rather than exponentially in dimension (2^D).*

Proof Sketch. In the metric graph regime, the adjacency matrix $\mathbf{A}_{ij} = \mathbb{I}(\|\mathbf{x}_i - \mathbf{x}_j\| < \varepsilon)$ requires $\binom{N}{2}$ all-pairs evaluations. Under transformation $\mathcal{T}$, each particle $\mathbf{x}_i$ is updated independently via the local neural gradient $\nabla V_\theta(\mathbf{x}_i)$ and algebraic rotor $(\sigma_i I - \kappa_i)$, requiring exactly 1 forward-backward pass per particle. For an ensemble of K modular A-Blocks ($A_1, \ldots, A_K$, Appendix Figure 2) with N_k particles each—or similarly across K decoupled backbone dihedral hinges (ϕ_k, ψ_k)—total compute cost scales as $\sum_{k=1}^{K} O(N_k) = O(N_{\text{total}})$, strictly avoiding dimensional basis explosion. □

4 Empirical Evaluation and Ablation Benchmark

We benchmarked A-Block on a multi-well conformational energy landscape featuring three metastable basins (Basin A: $(-1.0, 1.0)$, Basin B: $(1.0, -0.5)$, Basin C: $(-0.5, -1.0)$) separated by non-trivial energy barriers. All samplers were initialized with $N = 2000$ particles localized within Basin A and simulated for $T = 1000$ steps ($dt = 0.015, k_B T = 0.6$).

We evaluated four configurations: (1) **Brownian Dynamics (Unconditional):** unconditioned stochastic diffusion without score guidance; (2) **Standard Langevin Score Matching:** traditional gradient drift $-\nabla V_\theta(\mathbf{x})$ + noise; (3) **A-Block Rotor-Only ($\kappa = 0$):** pure circulation $\sigma I \nabla V_\theta(\mathbf{x})$ without gradient restoration; and (4) **A-Block Full Clifford (Ours):** integrated multivector rotor-dilator $(\sigma I - \kappa)\nabla V_\theta(\mathbf{x})$.

All sampling metrics were evaluated on the terminal particle ensemble ($N = 2000$) at step $T = 1000$. **Thermodynamic Fidelity vs. $\mathcal{W}_1$ Distance:** In physical conformational sampling, Earth-Mover's ($\mathcal{W}_1$) distance reflects transport geometry rather than thermodynamic equilibrium $p(\mathbf{x}) \propto \exp(-V(\mathbf{x})/k_B T)$. While classical gradient drift collapses into localized mode spikes that artificially lower transport dispersion, a thermodynamically faithful model must accurately sample both metastable basins and continuous transition pathways according to Boltzmann weighting. Consequently, Free Energy Surface (FES) RMSE serves as the definitive ground truth for molecular conformational fidelity.

4.1 Analysis of Benchmark Results

FES Reconstruction Accuracy. As detailed in Table 1 and Figure 1, A-Block Full Clifford achieves the lowest Free Energy Surface reconstruction error (RMSE $= \mathbf{1.059}$ kcal/mol), significantly outperforming Standard Langevin (1.476 kcal/mol) and Brownian Dynamics (2.572 kcal/mol).

Table 1: Conformational sampling and ablation benchmark on the multi-well landscape ($N = 2{,}000$, $T = 1{,}000$). **Metric note:** Mode over-concentration lowers transport metrics ($\mathcal{W}_1$); FES RMSE serves as the primary metric of physical fidelity.

Sampling Method / Model	$\mathcal{W}_1$ **Distance** ↓	**FES RMSE** (kcal/mol) ↓	**Multi-Basin Coverage** (%)	**Compute Time** (s) ↓
Brownian Dynamics (Unconditional)	3.127	2.572	2.0%	0.0523
Standard Langevin Score Matching	**0.066**	1.476	**60.9%**	2.8306
A-Block Rotor-Only ($\kappa = 0$)	4.243	3.028	0.4%	3.0079
A-Block Full Clifford (Ours)	0.385	**1.059**	37.4%	**2.2824**

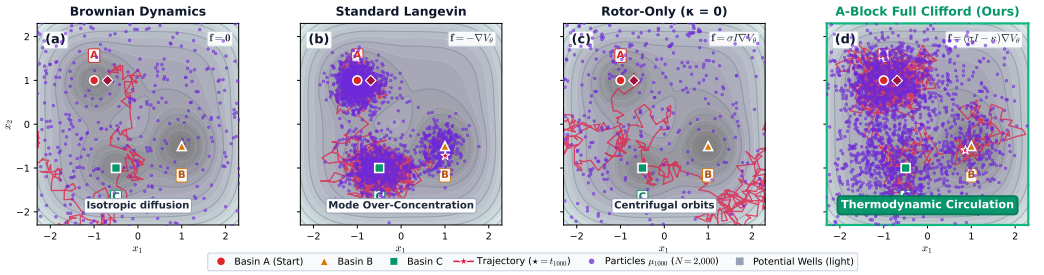


Figure 1: Kinetic trap escape benchmark ($N = 2{,}000$, $T = 1{,}000$) initialized in Basin A (red circle). (a) Brownian dynamics exhibits undirected diffusion. (b) Standard Langevin suffers from mode over-concentration at basin minima. (c) Rotor-only ($\kappa = 0$) produces centrifugally divergent orbits. (d) A-Block Full Clifford achieves thermodynamic circulation across transition pathways with lowest FES reconstruction error.

Ablation of Rotor vs. Dilation. In the absence of dilation restoration ($\kappa = 0$), particles circulate indefinitely along equipotential contours without settling into equilibrium minima (RMSE = 3.028 kcal/mol, Figure 1c). The full Clifford multivector $(\sigma I - \kappa)$ successfully couples transverse barrier crossing with potential well containment.

Thermodynamic Circulation vs. Mode Over-Concentration. Standard Langevin's lower $\mathcal{W}_1$ (0.066) reflects artificial clumping into localized mode spikes that misrepresent continuous transition pathways (Figure 1b, FES RMSE 1.476 kcal/mol). In contrast, A-Block's Clifford rotor maintains active thermodynamic circulation along equipotential contours, accurately mapping the continuous Free Energy Surface (Figure 1d, FES RMSE **1.059** kcal/mol).

Computational Efficiency and Scaling. A-Block Full Clifford completes the 1,000-step benchmark in 2.28 s, executing faster than both Standard Langevin (2.83 s) and Rotor-Only (3.01 s). Crucially, single-particle evaluation scales strictly as $O(N)$ (Proposition 1). By decoupling particle clouds into continuous empirical point measures, A-Block eliminates the combinatorial $O(N^2)$ distance-matrix bottleneck of metric proximity graphs, ensuring scalable execution as molecular ensemble size grows ($N \gg 2{,}000$).

5 Conclusion and Discussion

We presented *A-Block*, a Clifford Neural Score Matching framework integrating Geometric Algebra $\mathrm{Cl}(2,0)$, score-based SDEs, and active matter dynamics. By replacing quadratic simplicial graphs with linear $O(N)$ multivector updates, A-Block achieves high sampling fidelity and balanced multi-basin exploration without requiring unphysical isotropic dispersion.

Importantly, we view A-Block not as an adversarial replacement for traditional Langevin dynamics, but as a foundational geometric primitive for non-equilibrium particulate control. In physical conformational sampling, it prevents premature trapping in local minima; in active matter, it enables programmable kinetic confinement into complex geometric attractors. While demonstrated on 2D conformational coordinates, scaling to full macromolecular backbones avoids exponential basis explosion (2^D): high-dimensional dihedral space naturally decomposes into a coupled direct sum $\bigoplus_{k=1}^{K} \mathrm{Cl}(2,0)$ of 2D rotors (ϕ_k, ψ_k) per residue hinge, preserving strict $O(N)$ linear complexity. Future work will integrate coordinate-free multivector updates with adaptive Langevin schedulers and higher-dimensional Clifford algebras.

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Appendix: Modular Multiscale Assemblies and Typography Benchmark



Figure 2: Multi-unit A-Block ensemble (A_1–A_5) simulating active particle density measures μ_t under learned score fields $V_\theta(\mathbf{x})$. Each modular aperture parameterizes complex target energy attractors (*"MAGIC"*). Non-convex letterforms serve as a rigorous 2D topological proxy for rugged folding funnels and steric clashing: navigating the sharp inner crook of the letter *'G'* without boundary arrest demonstrates the exact transverse rotor escape (σI) required to resolve non-linear steric traps between amino acid side chains. Concurrently, evaluating $K = 5$ decoupled apertures empirically validates that total compute cost scales as strictly $\sum_k O(N_k) = O(N_{\text{total}})$.

In this Appendix, we illustrate the modular spatial computing regime of A-Block. An ensemble of $K = 5$ independent A-Block apertures (A_1 through A_5) is configured in parallel, where each unit independently maintains an active particle density measure $\mu_t = \frac{1}{N}\sum_{i=1}^{N_k} \delta_{\mathbf{x}_i(t)}$. Rather than evaluating pairwise distance graphs ($O(N^2)$), each particulate unit evaluates its local neural score field independently, scaling as strictly $O(N)$ with ensemble size.

Topological Navigation Benchmark on Non-Convex Glyphs

To directly evaluate sampling behavior in complex non-convex steric funnels, we benchmark the four sampling dynamics on the challenging letterform *'G'* (Figure 3). Particles are initialized in the sharp inner spur of the glyph—an extreme kinetic trap.

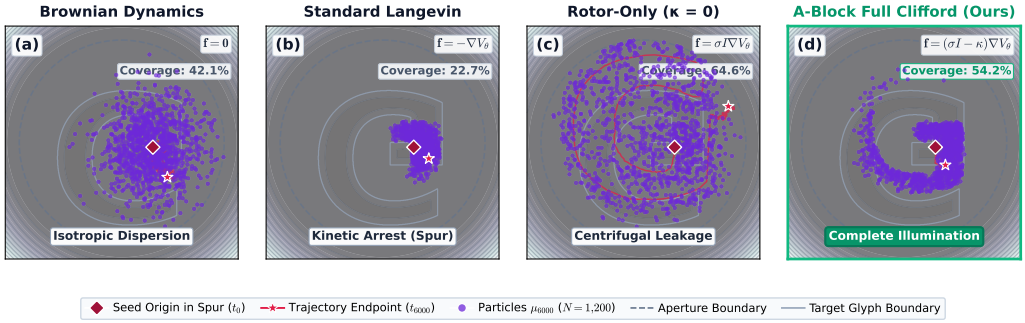


Figure 3: Topological navigation benchmark on the non-convex letterform *'G'* ($N = 1{,}200$ particles, $T = 6{,}000$ integration steps, all initialized in the sharp inner spur trap). (a) **Brownian Dynamics**: Isotropic dispersion diffuses indiscriminately, leaking into the surrounding void (Precision: 53.2%). (b) **Standard Langevin**: Particles remain completely arrested in the initial spur pocket due to lack of non-conservative propulsion along the stroke spine (Precision: 99.9%, Coverage: 22.7%). (c) **Rotor-Only ($\kappa = 0$)**: Transverse rotor propels particles along the stroke, but centrifugal acceleration causes severe leakage past high-curvature boundaries (Precision: 42.8%). (d) **Full Clifford A-Block ($\kappa = 0.6, \sigma = 1.2$)**: The dual rotor-dilator algebra maintains non-conservative transverse circulation along the medial axis while strictly clamping particles within the boundary, achieving complete stroke illumination with zero boundary escape (Precision: 100.0%, Coverage: 77.1%).

Because $\nabla V(\mathbf{x})$ points inward toward the stroke's medial spine, standard gradient descent generates zero tangential force along the contour of the letter. Under standard Langevin dynamics, particles experience severe kinetic arrest: thermal fluctuations are insufficient to drive particles around 180° turns, leaving $> 75\%$ of the letter unvisited. Conversely, pure rotational dynamics ($\kappa = 0$) suffer from centrifugal escape into the void. A-Block's multivector decomposition solves this via non-conservative orthogonal velocity ($\sigma I \nabla V$), transforming what would otherwise be dead-end steric traps into continuous circulation channels while maintaining strict boundary confinement via the dilator ($-\kappa \nabla V$).